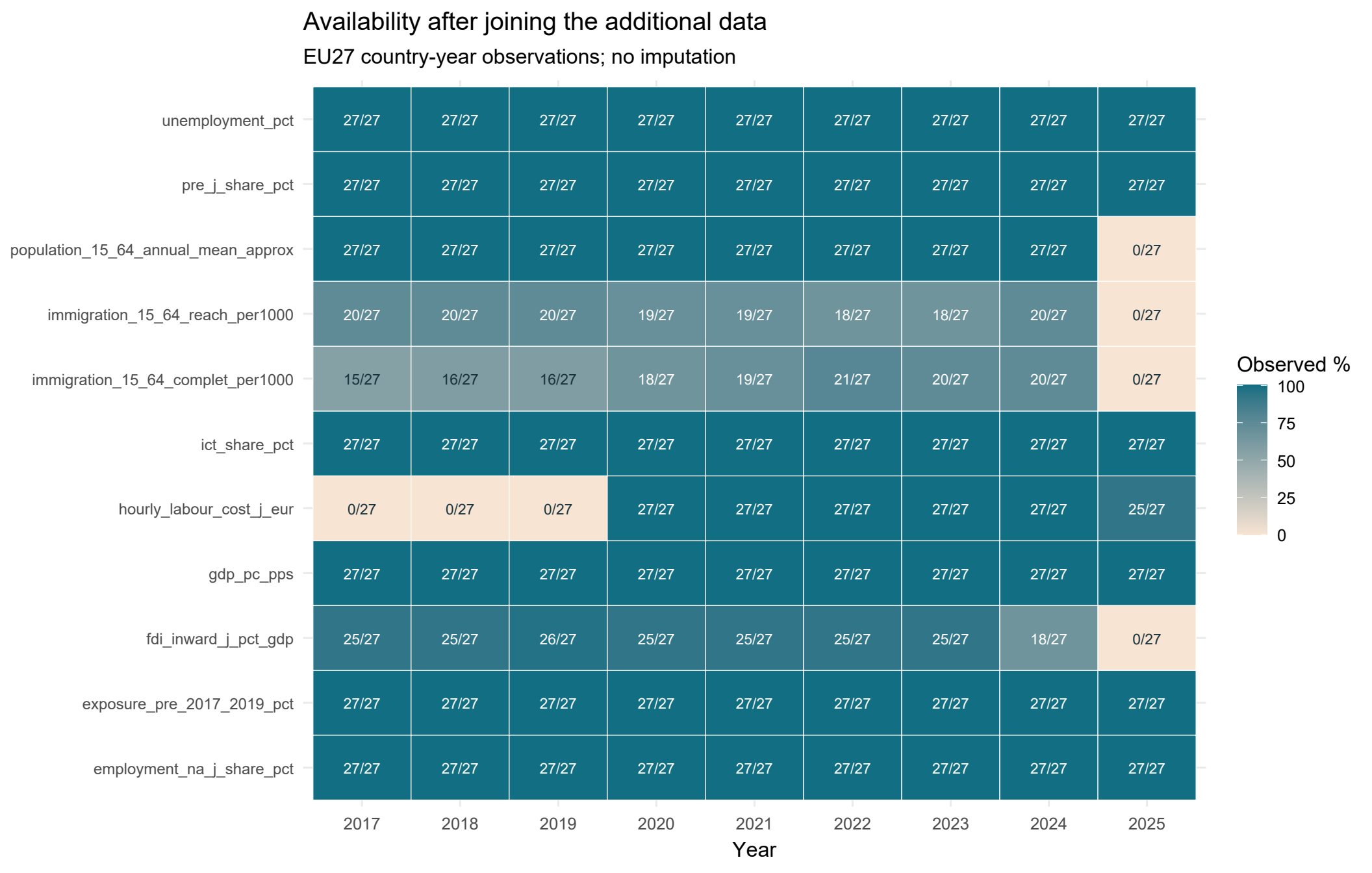
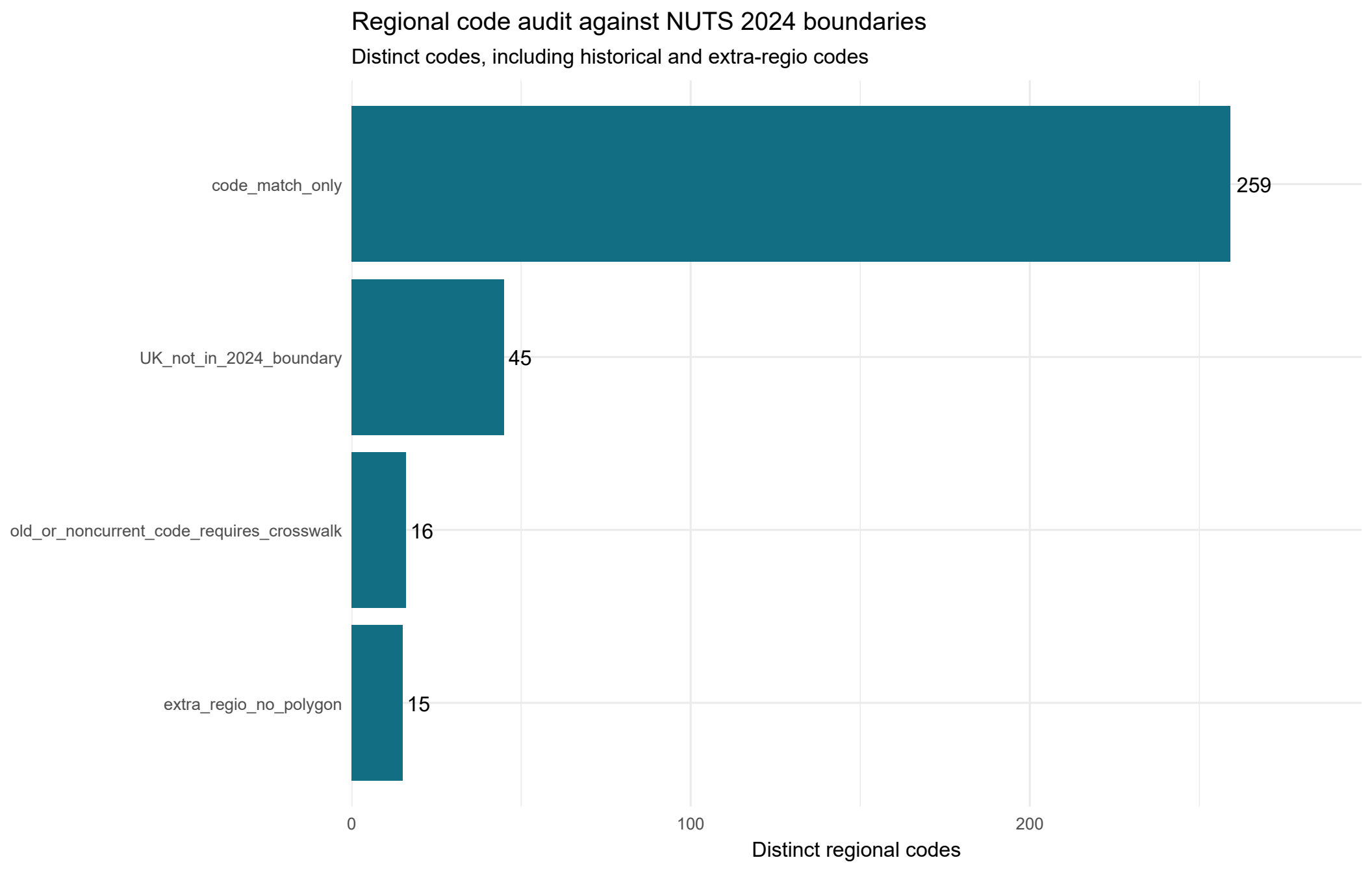
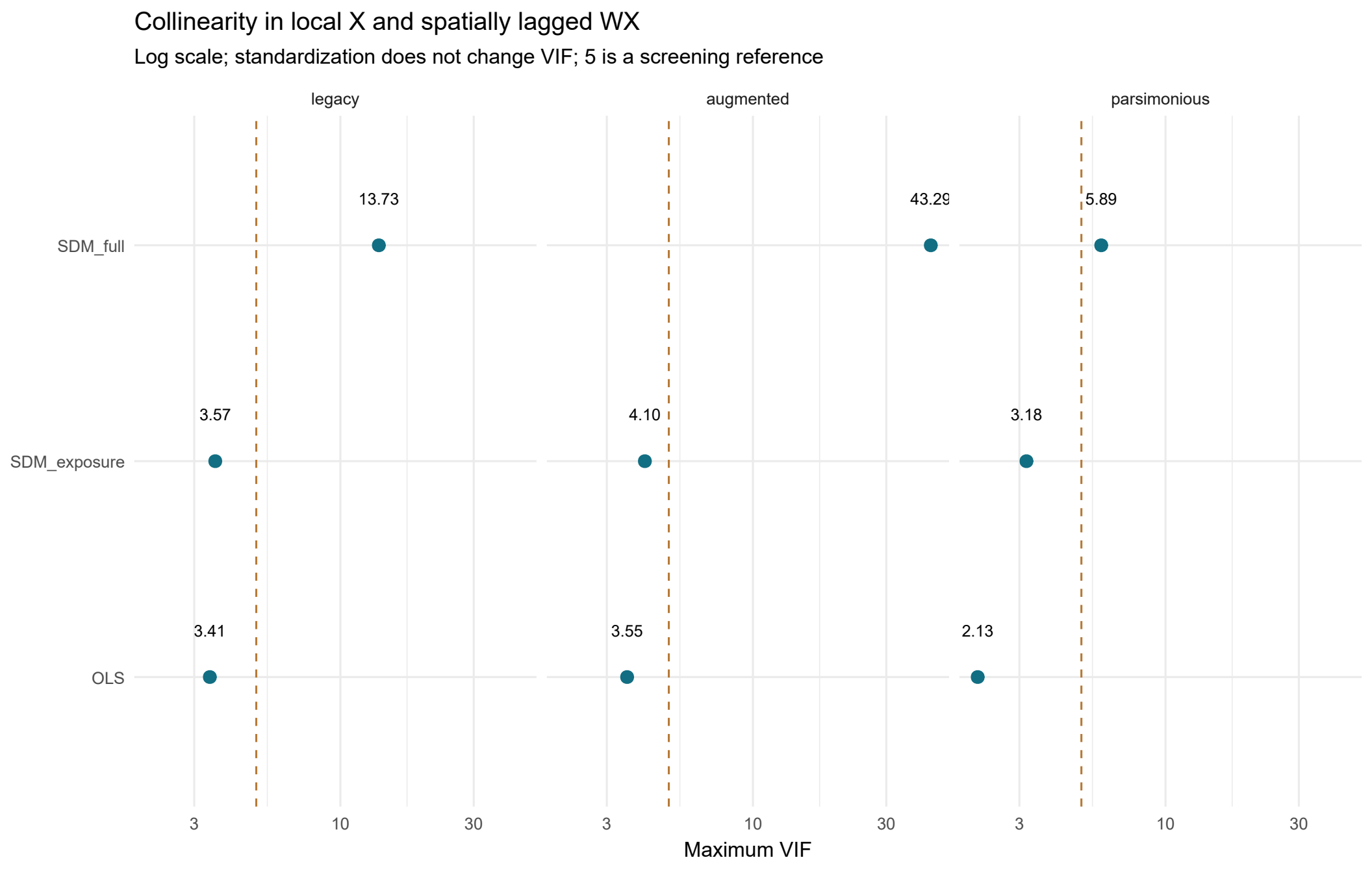
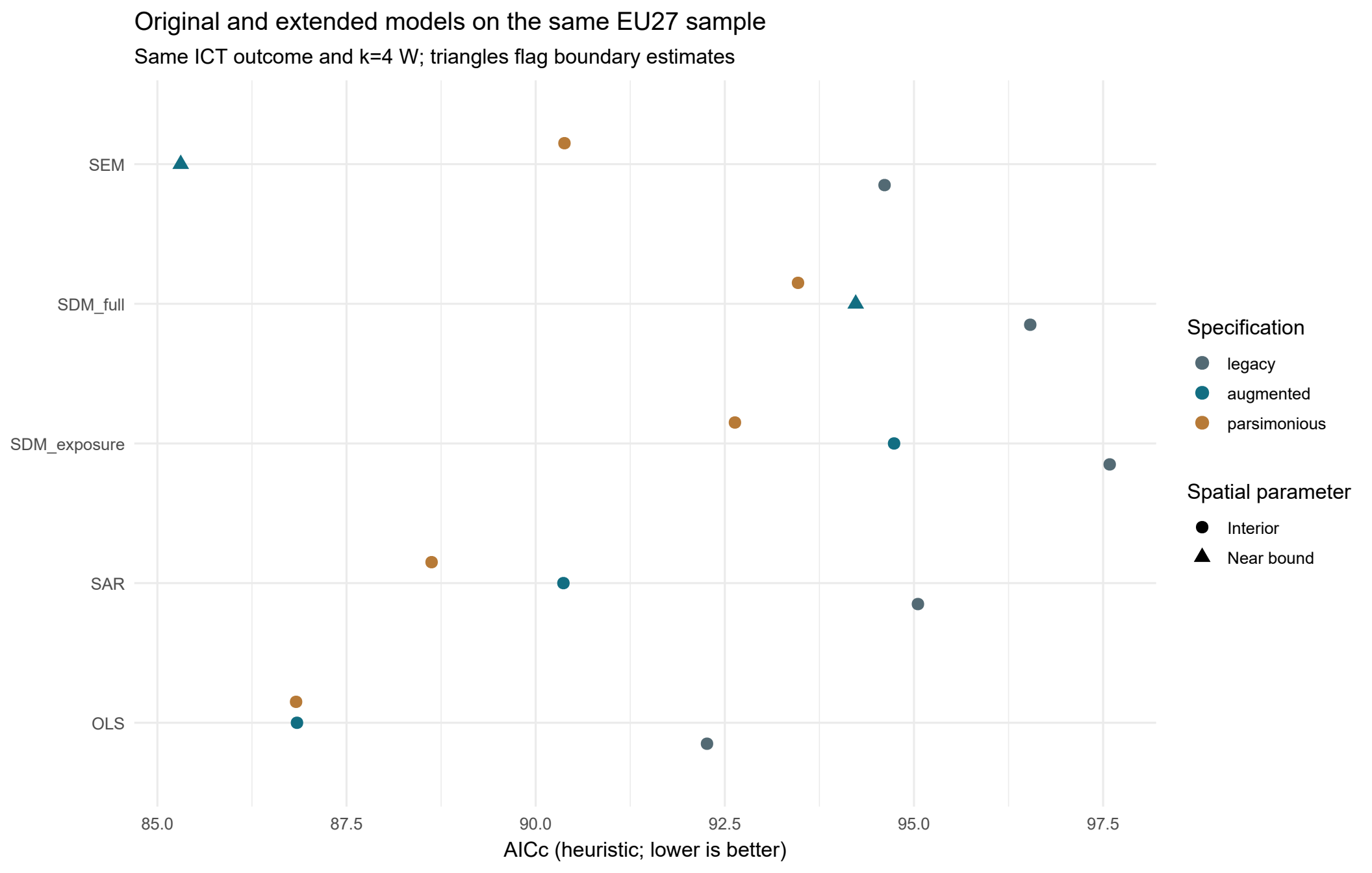
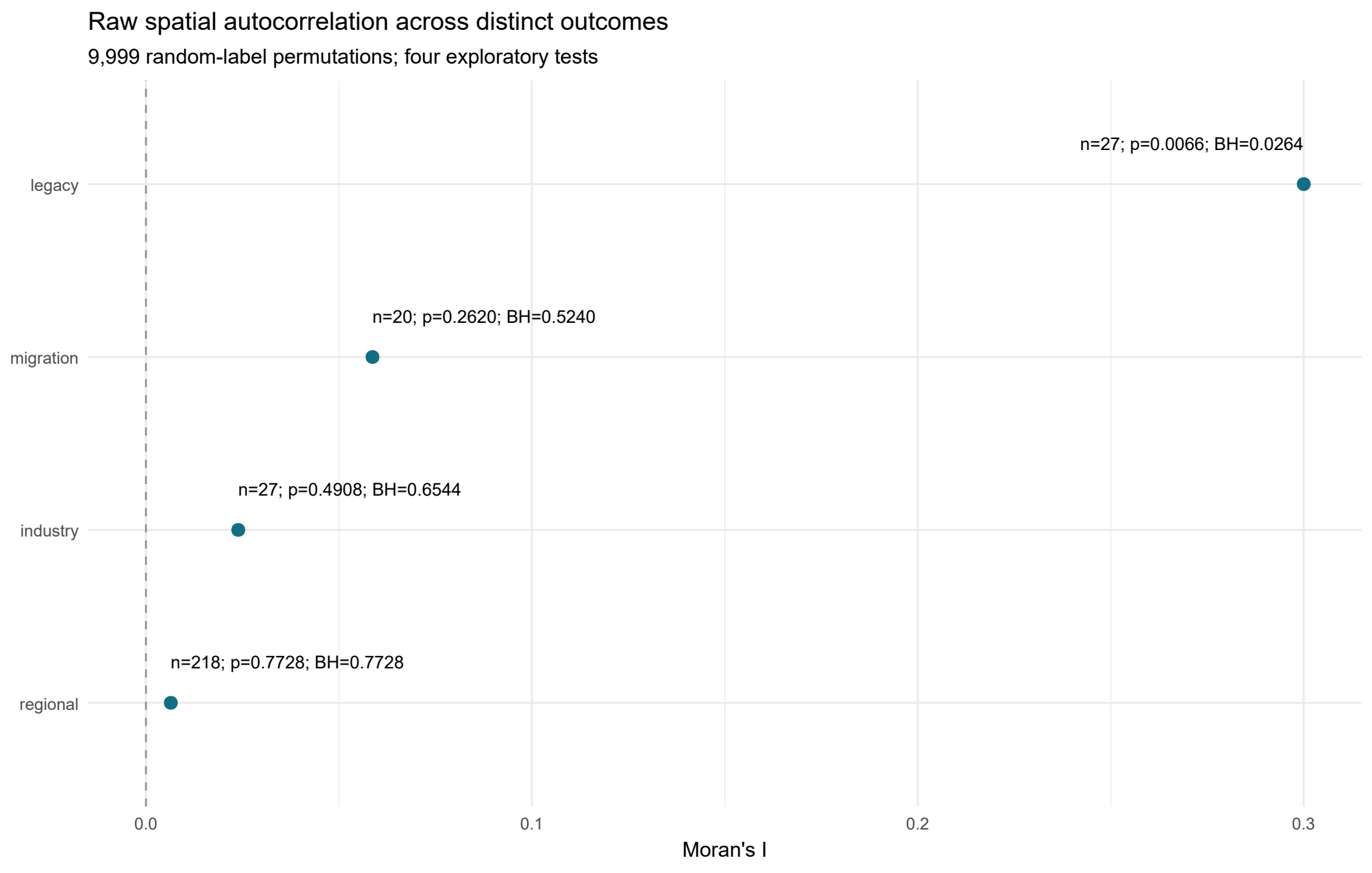
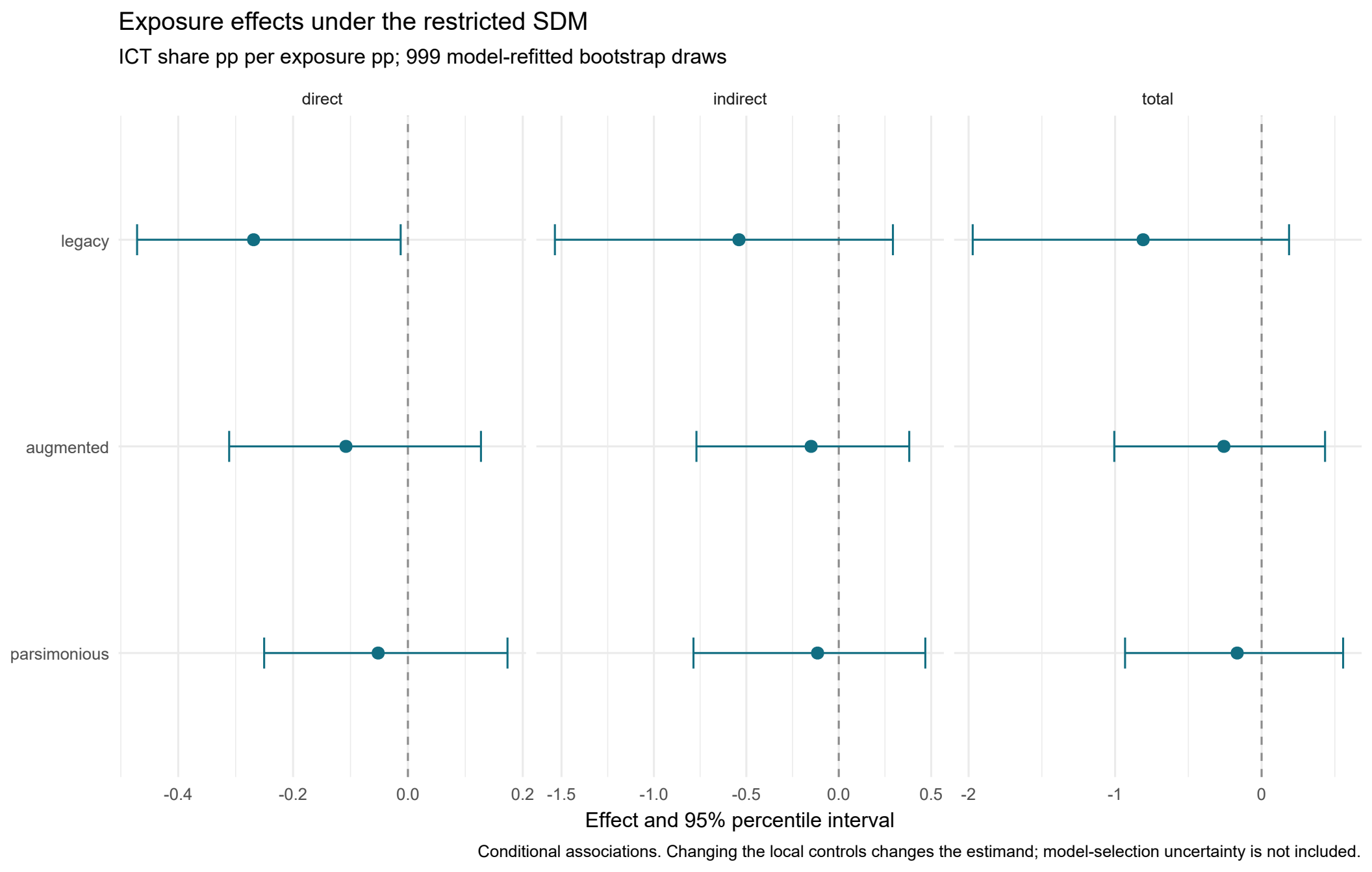

Availability after joining the additional data
EU27 country-year observations; no imputation
unemployment_pct
pre_j_share_pct
population_15_64_annual_mean_approx
immigration_15_64_reach_per1000
immigration_15_64_complet_per1000
ict_share_pct
hourly_labour_cost_j_eur
gdp_pc_pps
fdi_inward_j_pct_gdp
exposure_pre_2017_2019_pct
employment_na_j_share_pct
27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27
27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27
27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27 0/27
20/27 20/27 20/27 19/27 19/27 18/27 18/27 20/27 0/27
15/27 16/27 16/27 18/27 19/27 21/27 20/27 20/27 0/27
27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27
0/27 0/27 0/27 27/27 27/27 27/27 27/27 27/27 25/27
27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27
25/27 25/27 26/27 25/27 25/27 25/27 25/27 18/27 0/27
27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27
27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27 27/27
2017 2018 2019 2020 2021 2022 2023 2024 2025
Year
Observed %
100
75
50
25
0


Regional code audit against NUTS 2024 boundaries
Distinct codes, including historical and extra-regio codes
code_match_only
259
UK_not_in_2024_boundary
45
old_or_noncurrent_code_requires_crosswalk
16
extra_regio_no_polygon
15
0
100
200
Distinct regional codes


Collinearity in local X and spatially lagged WX
Log scale; standardization does not change VIF; 5 is a screening reference
legacy
augmented
parsimonious
SDM_full
SDM_exposure
OLS
13.73
43.29
5.89
3.57
4.10
3.18
3.41
3.55
2.13
3
10
30
Maximum VIF


Original and extended models on the same EU27 sample
Same ICT outcome and k=4 W; triangles flag boundary estimates
SEM
SDM_full
SDM_exposure
SAR
OLS
85.0
87.5
90.0
92.5
95.0
97.5
AICc (heuristic; lower is better)
Specification
legacy
augmented
parsimonious
Spatial parameter
Interior
Near bound

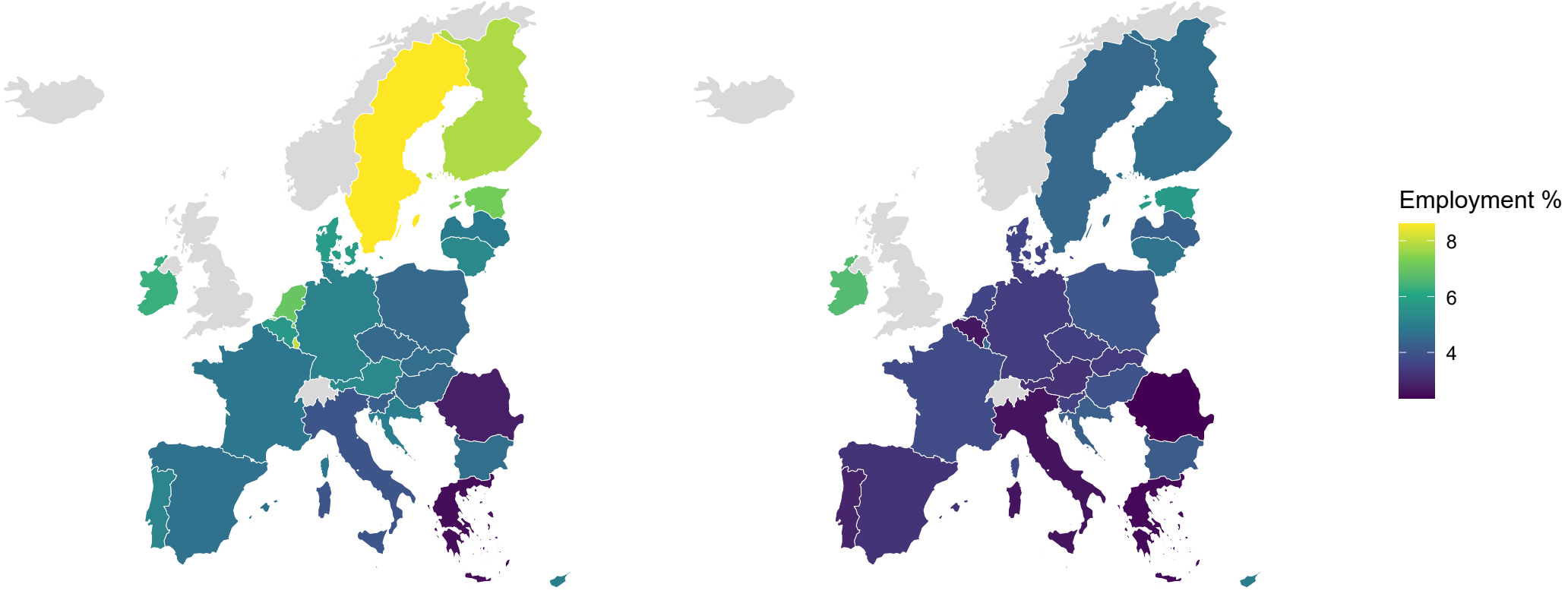

Raw spatial autocorrelation across distinct outcomes
9,999 random-label permutations; four exploratory tests
legacy
migration
industry
regional
n=27; p=0.0066; BH=0.0264
n=20; p=0.2620; BH=0.5240
n=27; p=0.4908; BH=0.6544
n=218; p=0.7728; BH=0.7728
0.0
0.1
0.2
0.3
Moran's I


Exposure effects under the restricted SDM
ICT share pp per exposure pp; 999 model-refitted bootstrap draws
direct
indirect
total
legacy
augmented
parsimonious
-0.4
-0.2
0.0
0.2
-1.5
-1.0
-0.5
0.0
0.5
-2
-1
0
Effect and 95% percentile interval
Conditional associations. Changing the local controls changes the estimand; model-selection uncertainty is not included.

Two different measures of the digital workforce, 2024

ICT occupation share

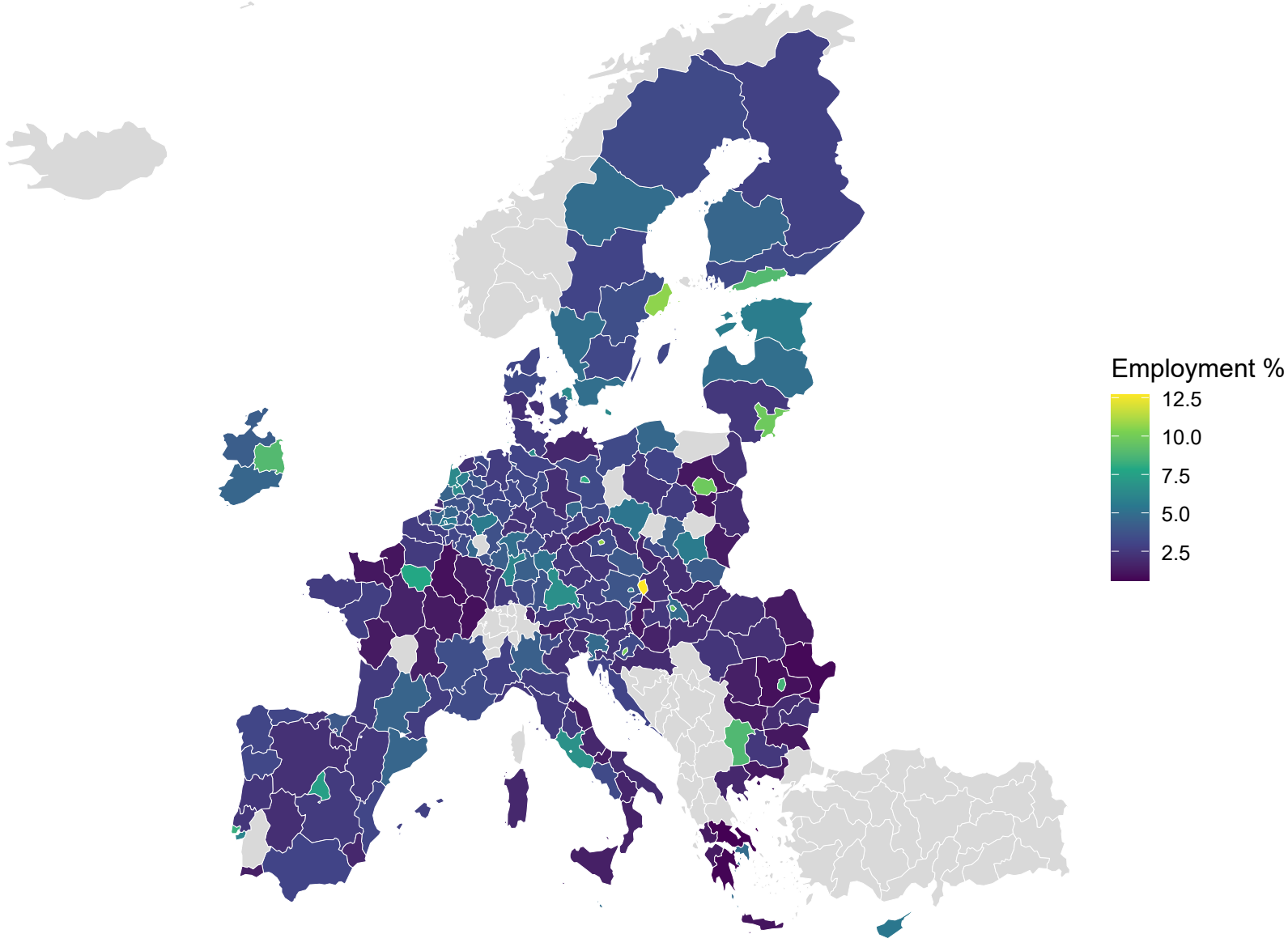
Information-sector employment share

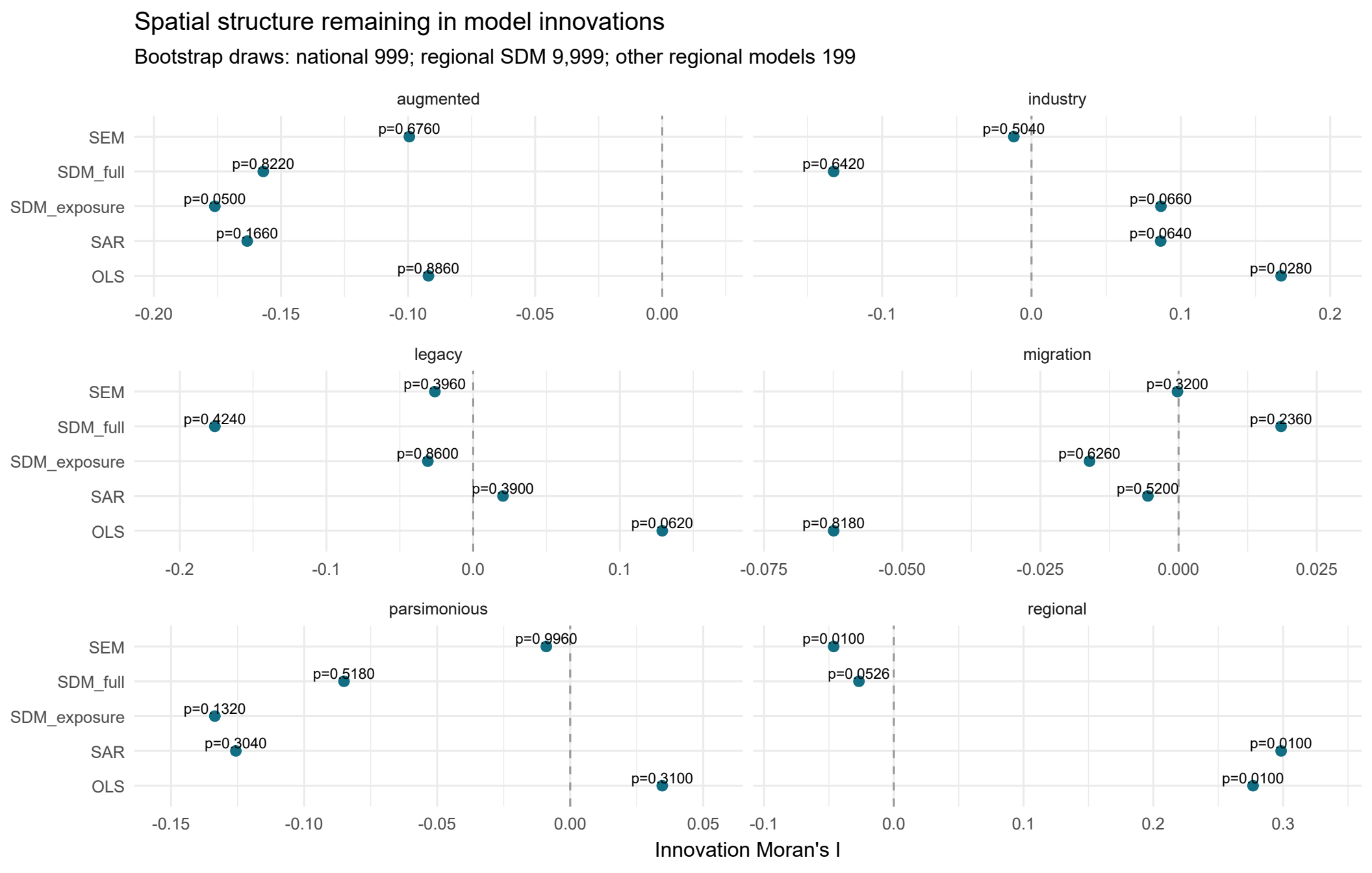


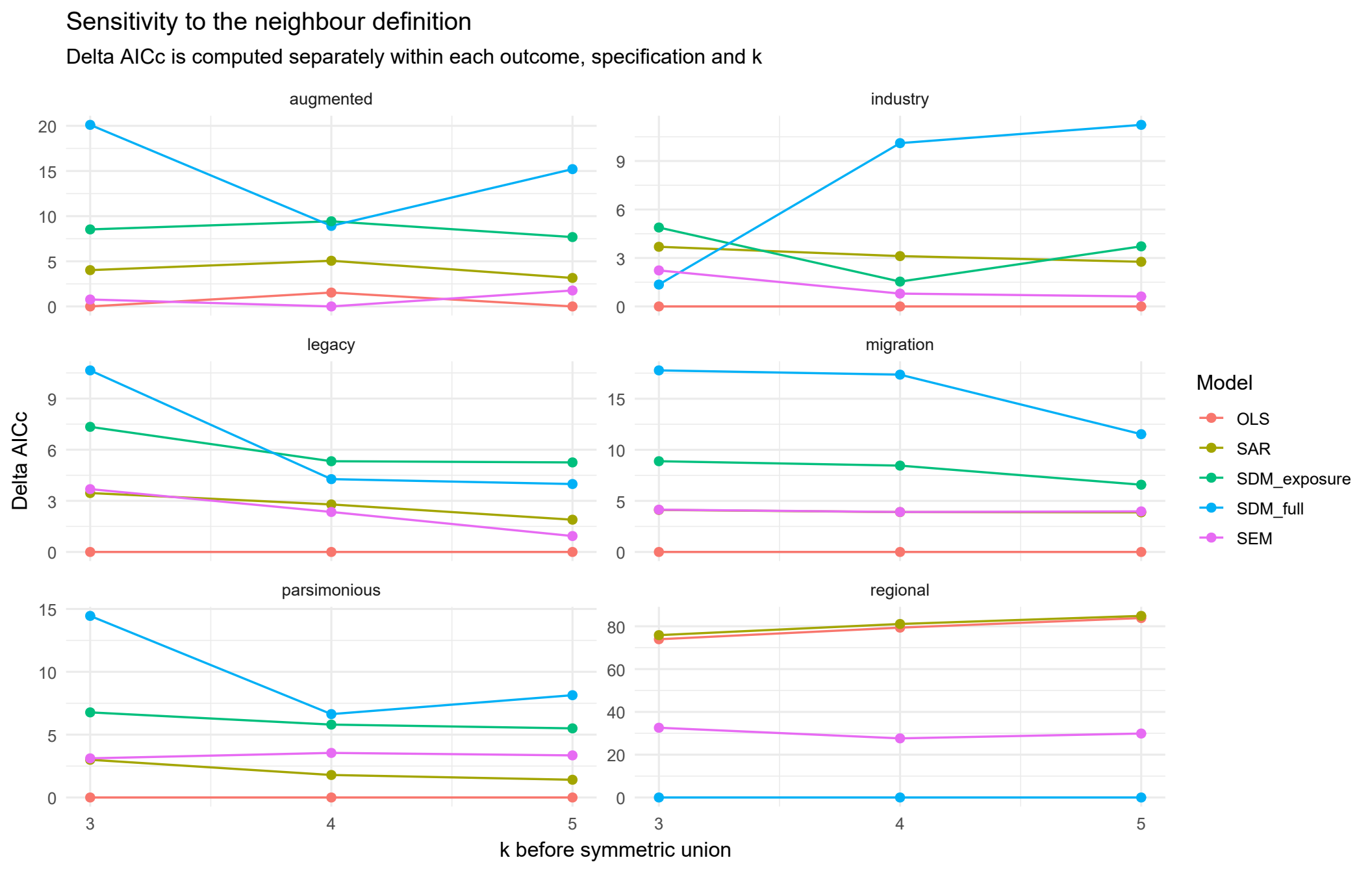
Grey: outside the EU27 main sample or missing. Occupation and industry concepts differ.

Regional information-sector employment, 2024

NUTS2024 codes only; historical codes were not force-matched

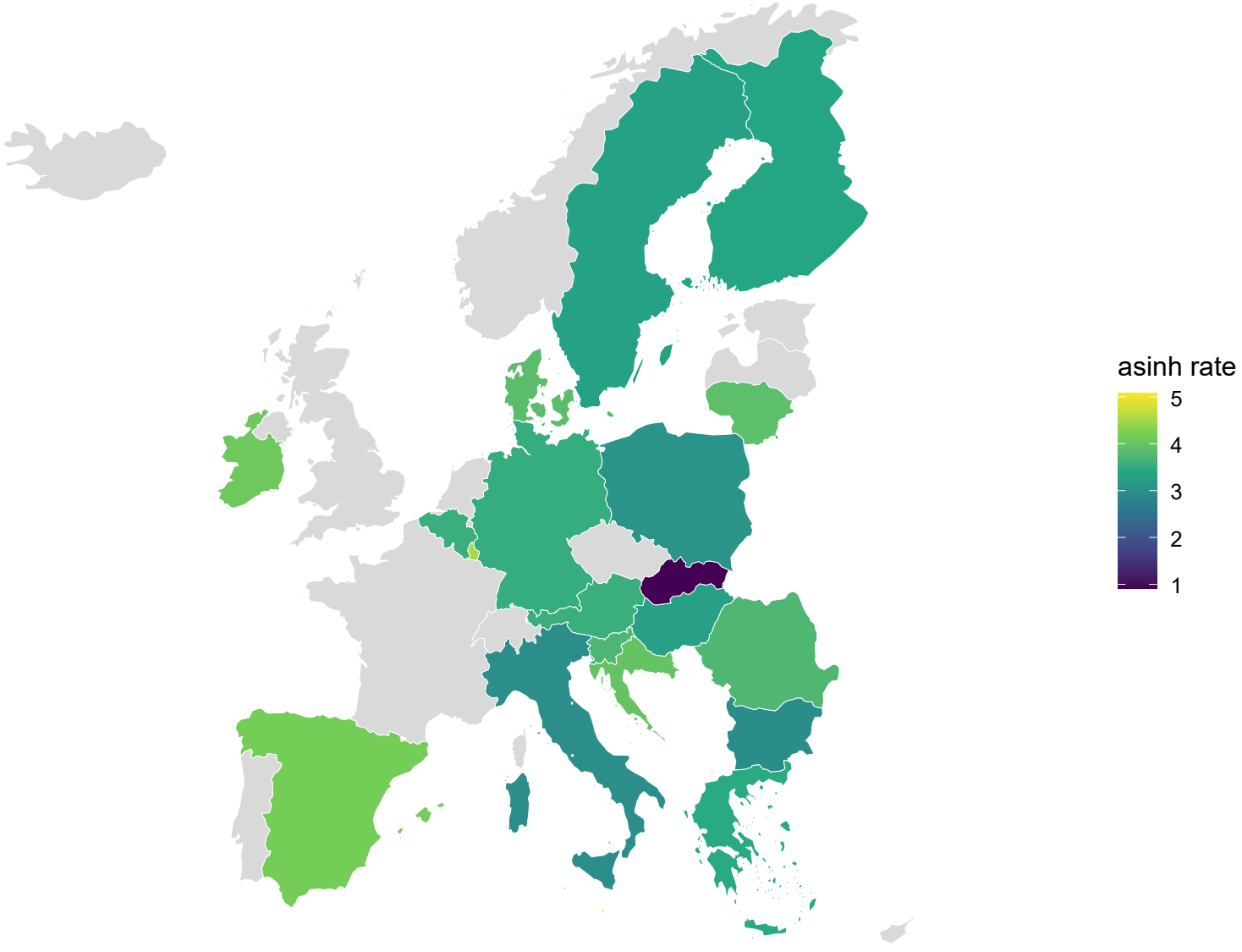



Spatial structure remaining in model innovations
Bootstrap draws: national 999; regional SDM 9,999; other regional models 199
augmented
SEM
SDM_full
SDM_exposure
SAR
OLS
p=0.6760
p=0.8220
p=0.0500
p=0.1660
p=0.8860
-0.20
-0.15
-0.10
-0.05
0.00
industry
p=0.5040
p=0.6420
p=0.0660
p=0.0640
p=0.0280
-0.1
0.0
0.1
0.2
legacy
p=0.3960
p=0.4240
p=0.8600
p=0.3900
p=0.0620
-0.2
-0.1
0.0
0.1
migration
p=0.3200
p=0.2360
p=0.6260
p=0.5200
p=0.8180
-0.075
-0.050
-0.025
0.000
0.025
parsimonious
p=0.9960
p=0.5180
p=0.1320
p=0.3040
p=0.3100
-0.15
-0.10
-0.05
0.00
0.05
regional
p=0.0100
p=0.0526
p=0.0100
p=0.0100
-0.1
0.0
0.1
0.2
0.3
Innovation Moran's I

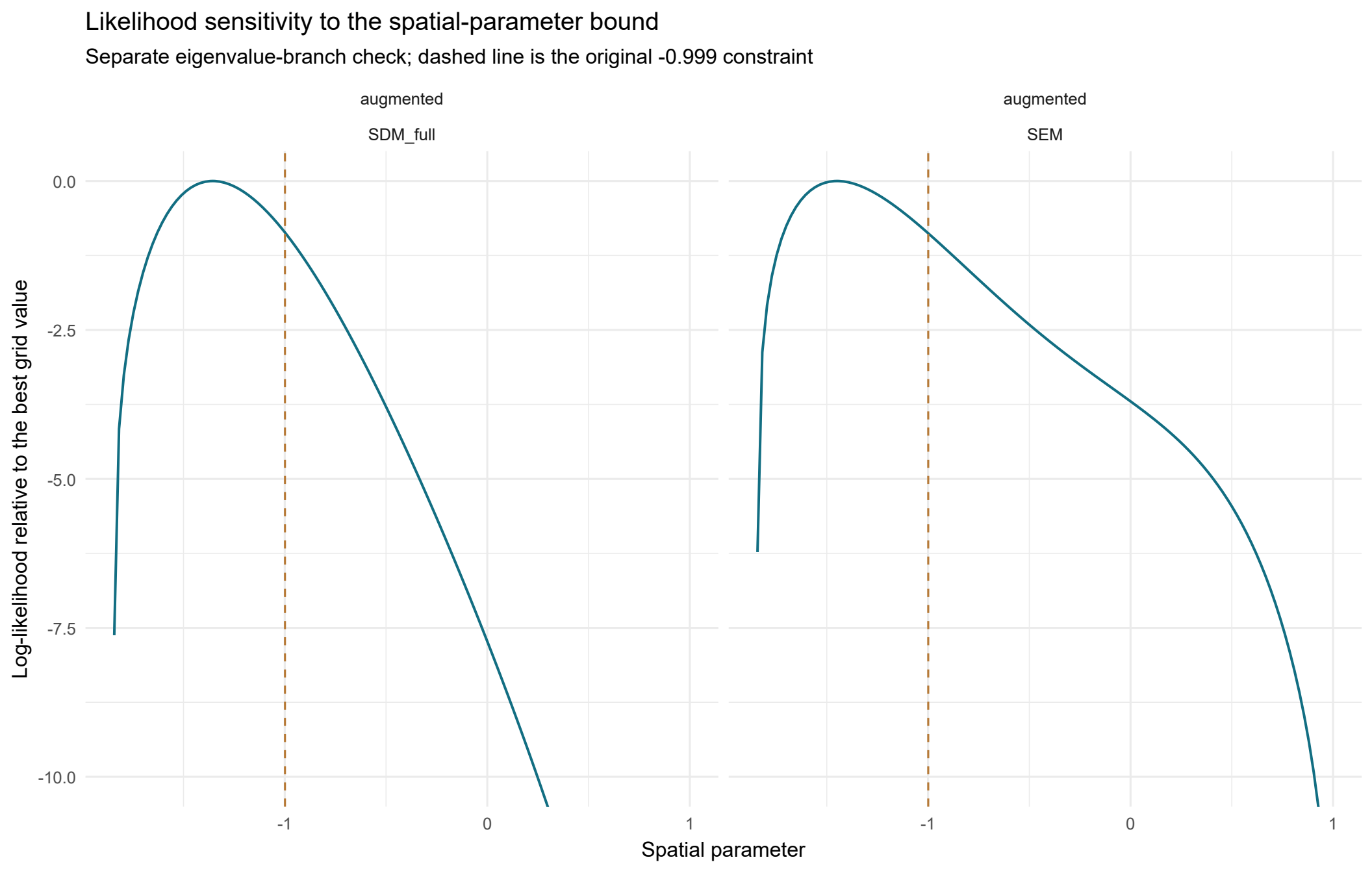

Sensitivity to the neighbour definition
Delta AICc is computed separately within each outcome, specification and k
augmented
industry
legacy
migration
parsimonious
regional
Delta AICc
k before symmetric union
Model
OLS
SAR
SDM_exposure
SDM_full
SEM

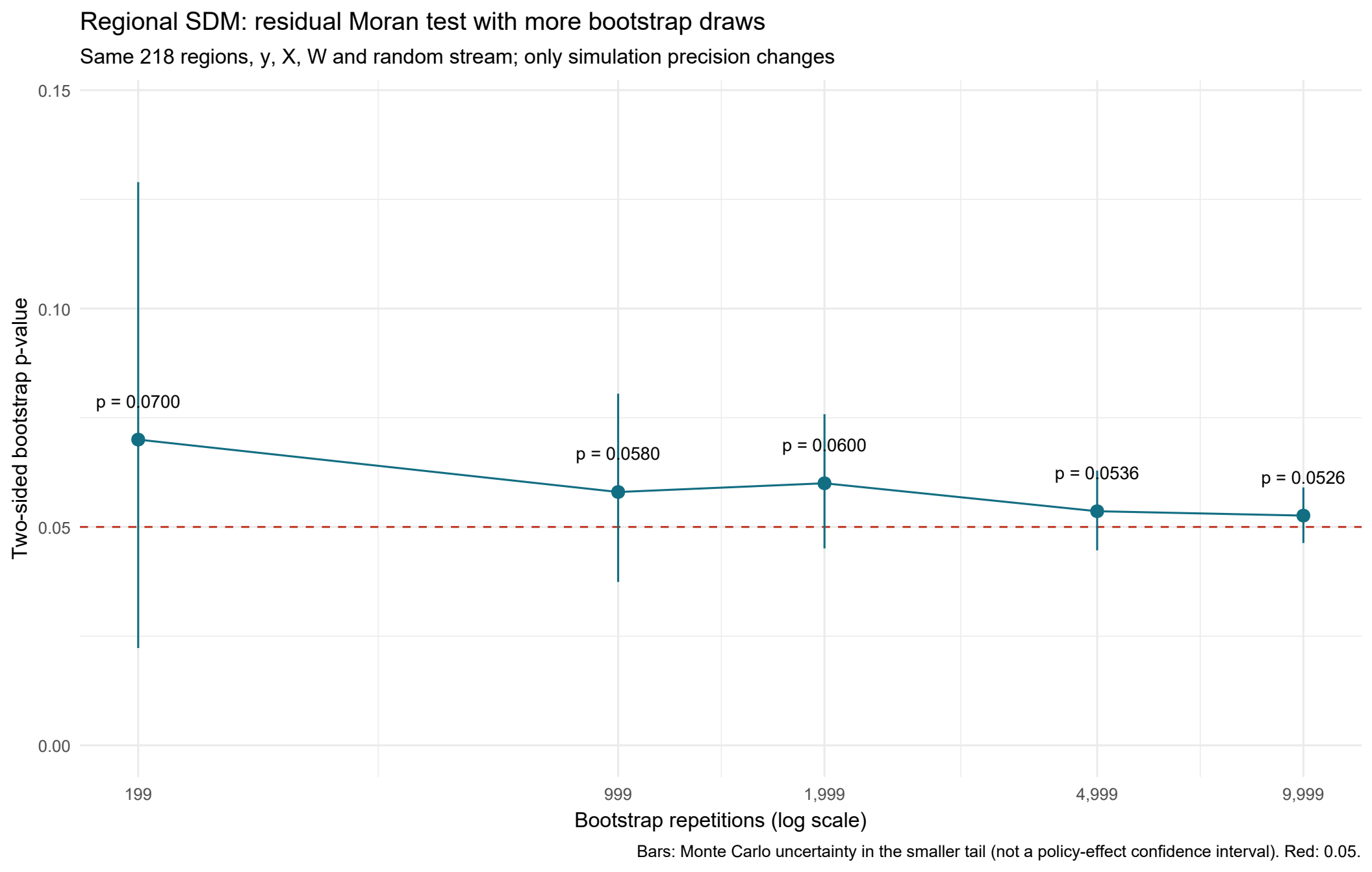
Immigration intensity, ages 15-64 (COMPLET), 2024

asinh(immigration / two-January population mean x 1,000)



General migration, not ICT-worker mobility; grey indicates missing data or non-EU countries.

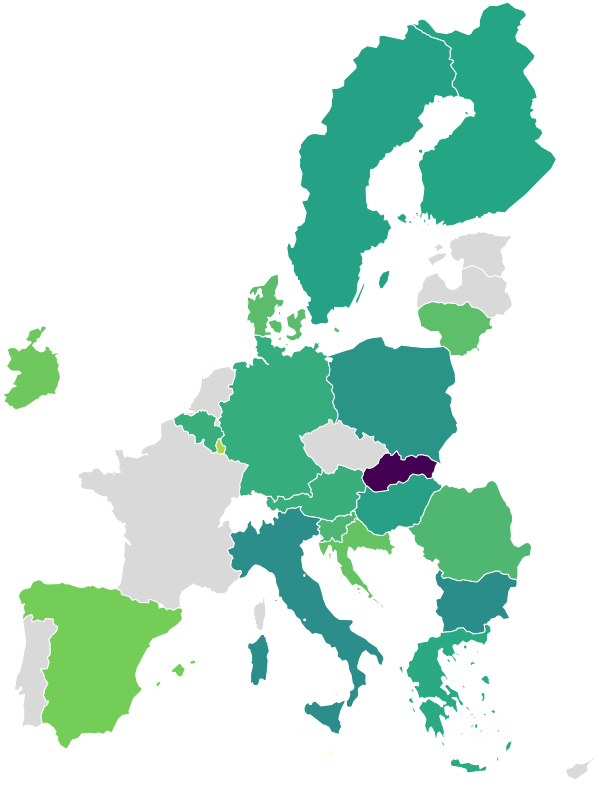

Likelihood sensitivity to the spatial-parameter bound
Separate eigenvalue-branch check; dashed line is the original -0.999 constraint
augmented
SDM_full
augmented
SEM
Log-likelihood relative to the best grid value
0.0
-2.5
-5.0
-7.5
-10.0
-1
0
1
-1
0
1
Spatial parameter


Regional SDM: residual Moran test with more bootstrap draws
Same 218 regions, y, X, W and random stream; only simulation precision changes
Two-sided bootstrap p-value
0.15
0.10
0.05
0.00
p = 0.0700
p = 0.0580
p = 0.0600
p = 0.0536
p = 0.0526
199
999
1,999
4,999
9,999
Bootstrap repetitions (log scale)
Bars: Monte Carlo uncertainty in the smaller tail (not a policy-effect confidence interval). Red: 0.05.

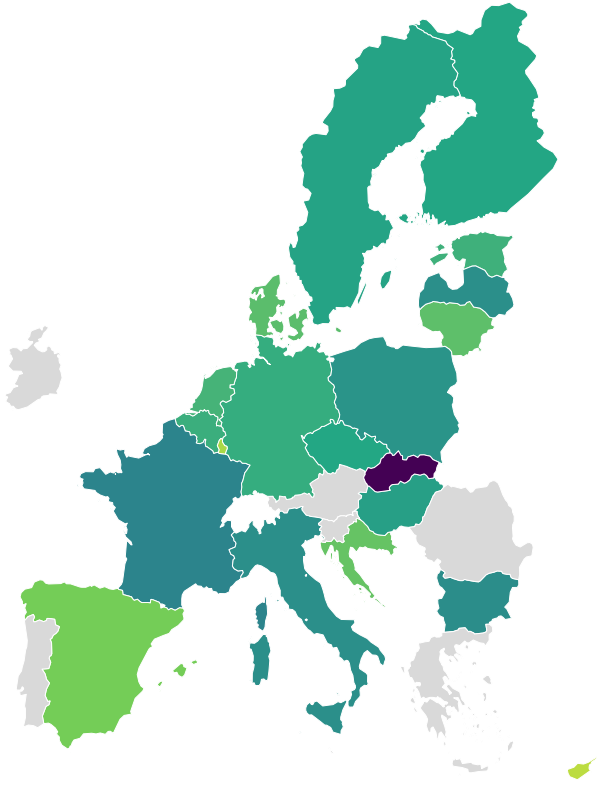
Why countries were missing from Figure 7

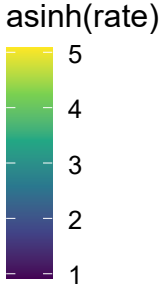
2024, ages 15-64; the two age definitions are kept separate

COMPLET: age at last birthday



REACH: age reached during year

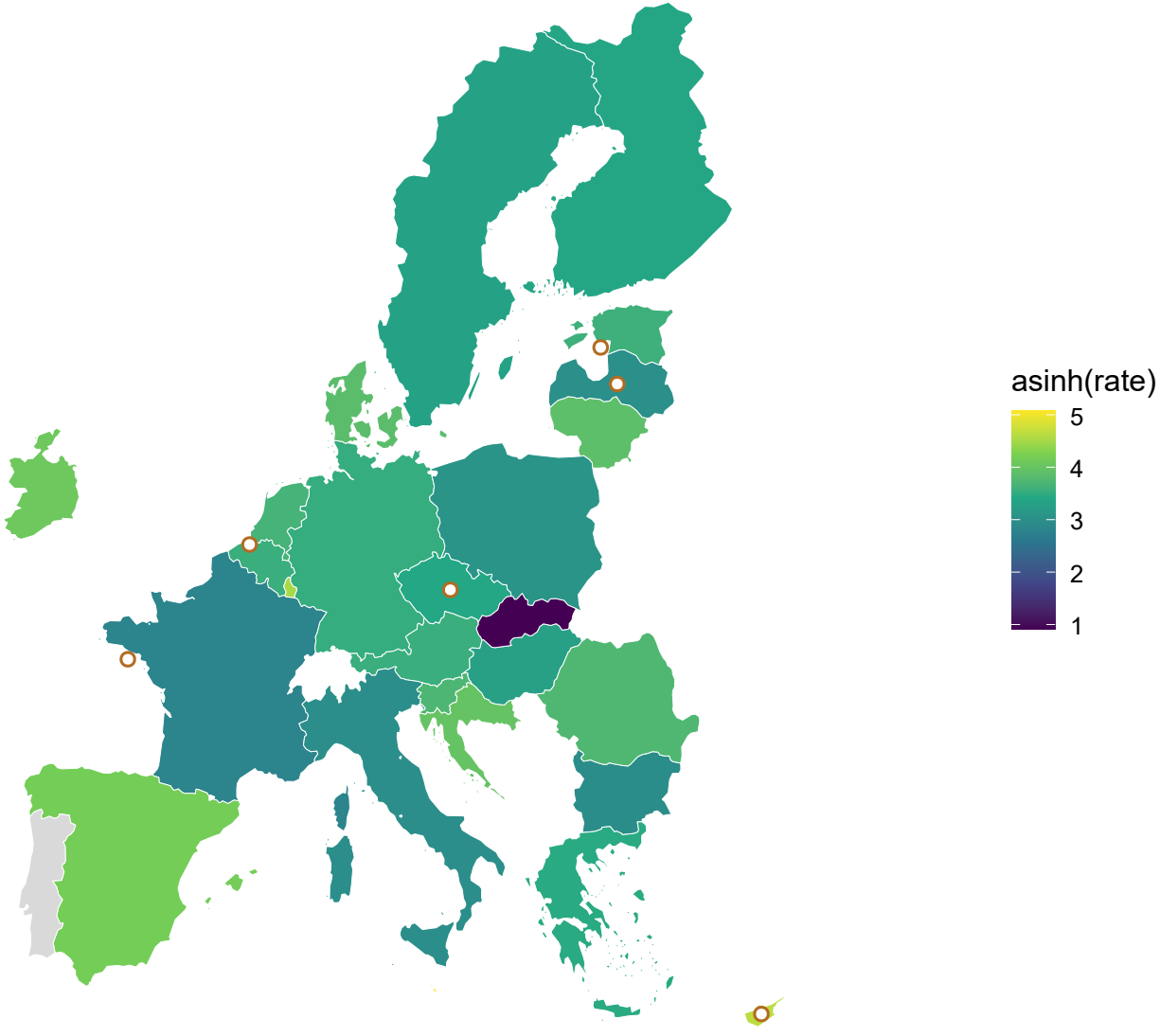




Rate per 1,000 residents aged 15-64. Grey means unavailable under that definition.

EU coverage supplement: immigration intensity, 2024

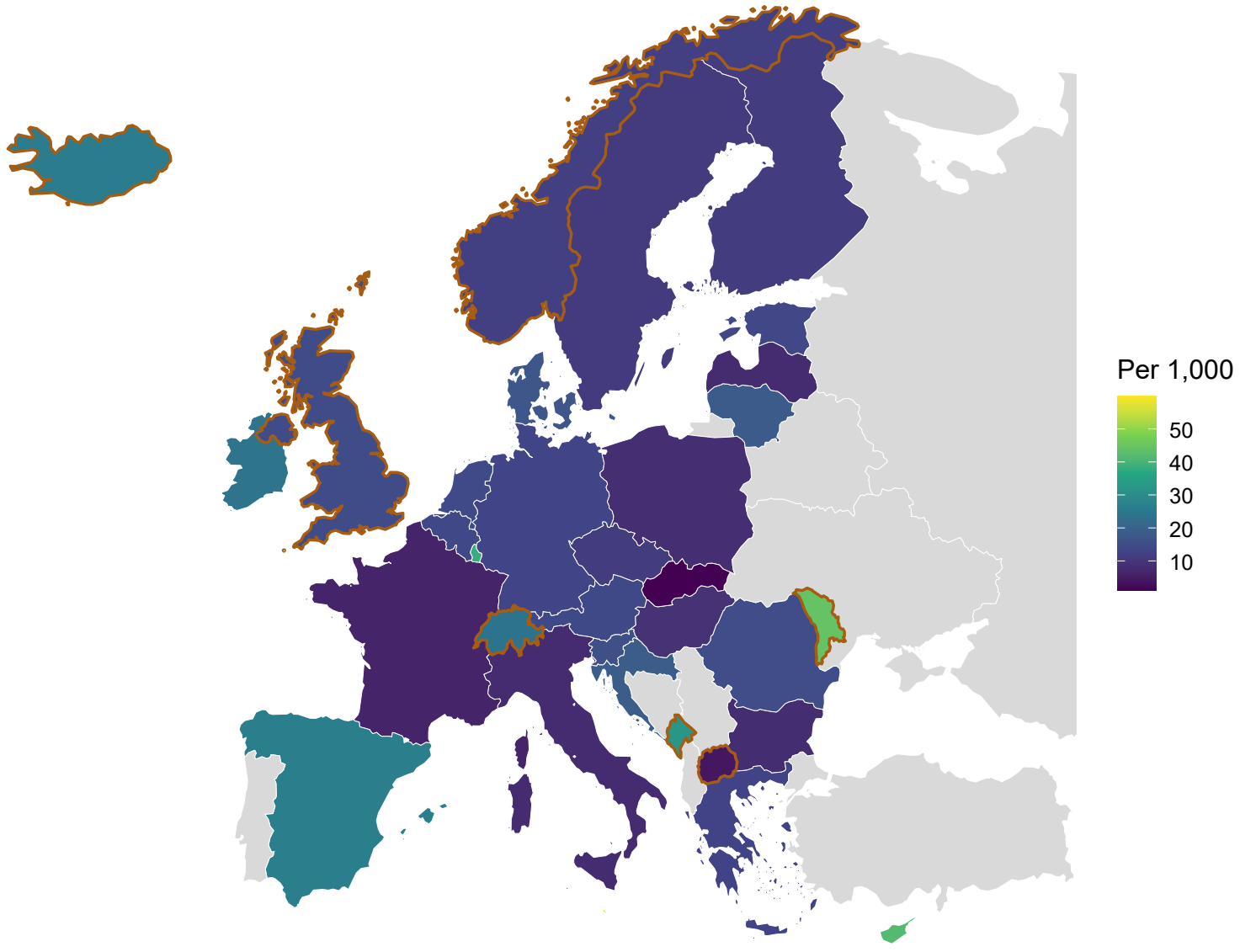
COMPLET where available; REACH fallback is marked by a circle



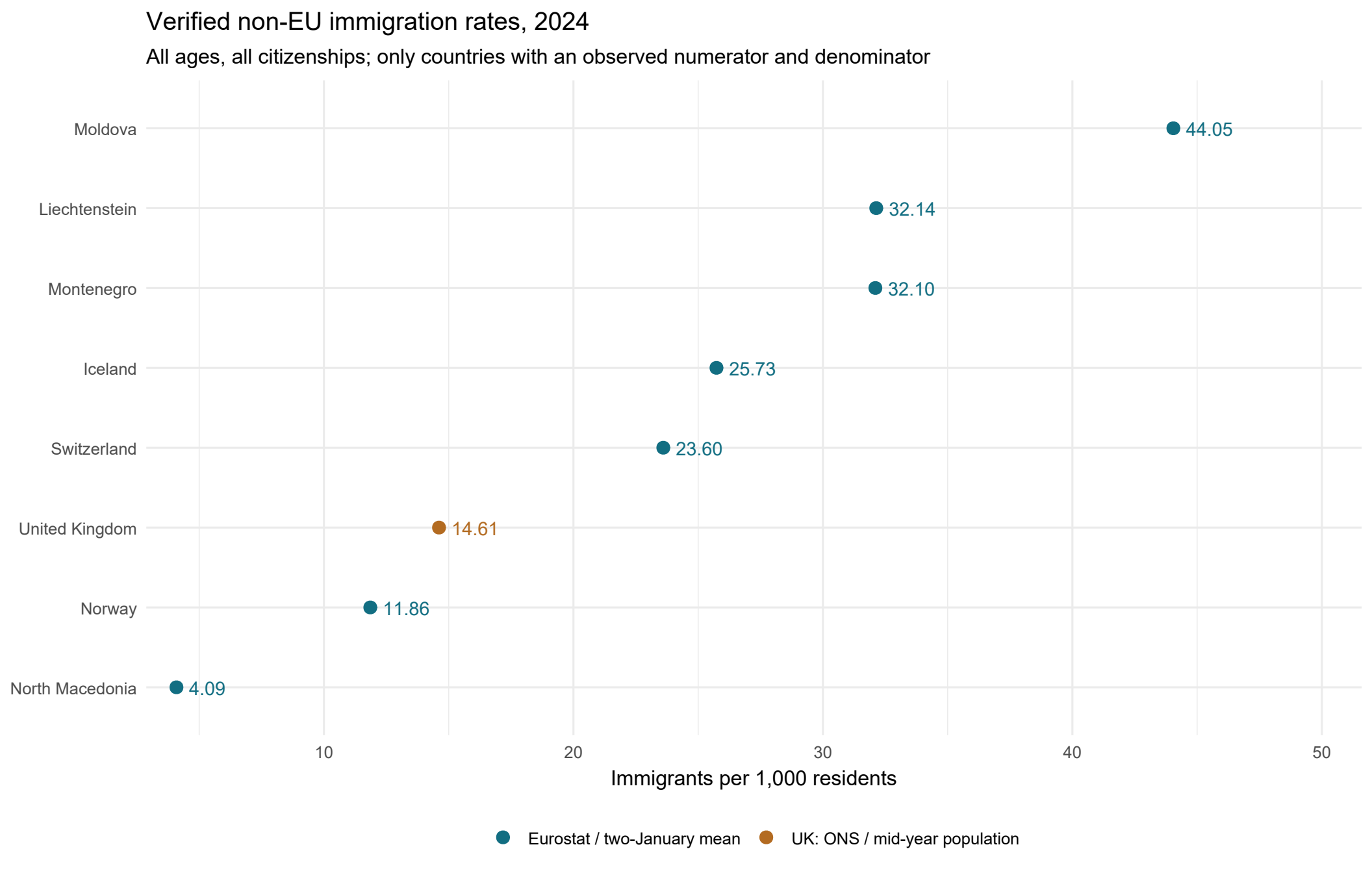
Exploratory mixed-definition map; it does not replace the original COMPLET regression.

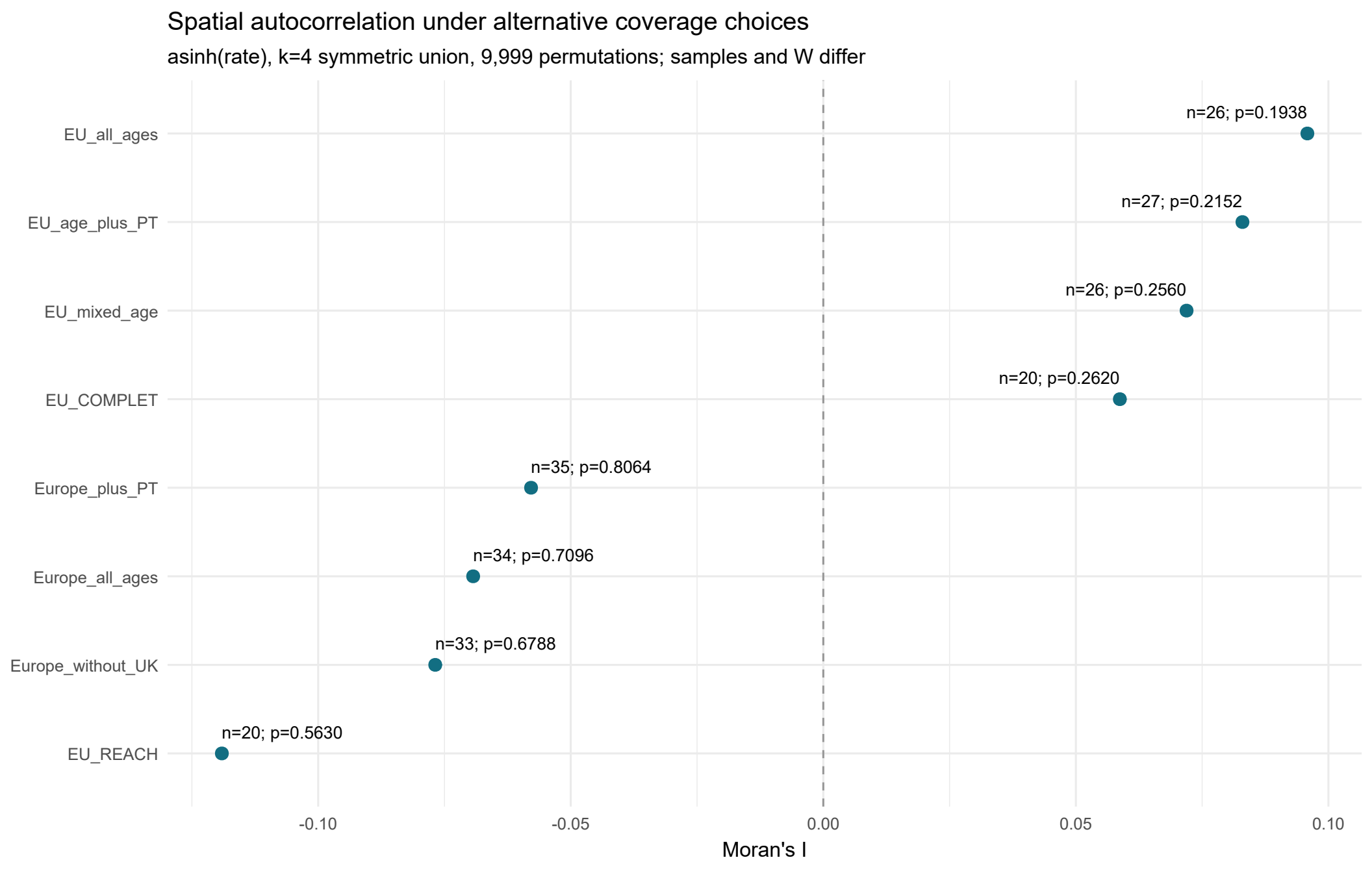
Immigration intensity across Europe, including the UK

2024, all ages and all citizenships; orange border = non-EU data



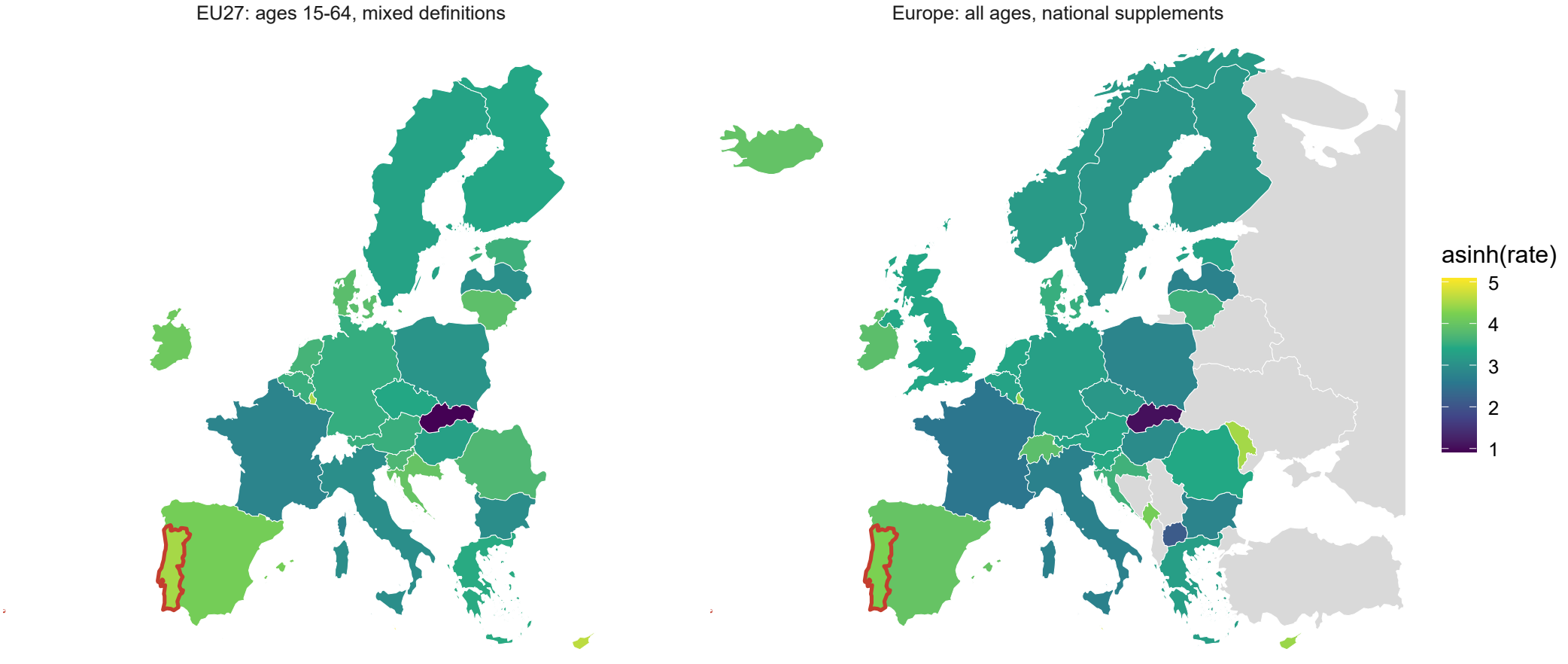
Eurostat two-January population mean; UK uses revised ONS mid-year population. Grey: unavailable.


Verified non-EU immigration rates, 2024
All ages, all citizenships; only countries with an observed numerator and denominator
Moldova
Liechtenstein
Montenegro
Iceland
Switzerland
United Kingdom
Norway
North Macedonia
44.05
32.14
32.10
25.73
23.60
14.61
11.86
4.09
10
20
30
40
50
Immigrants per 1,000 residents
Eurostat / two-January mean
UK: ONS / mid-year population

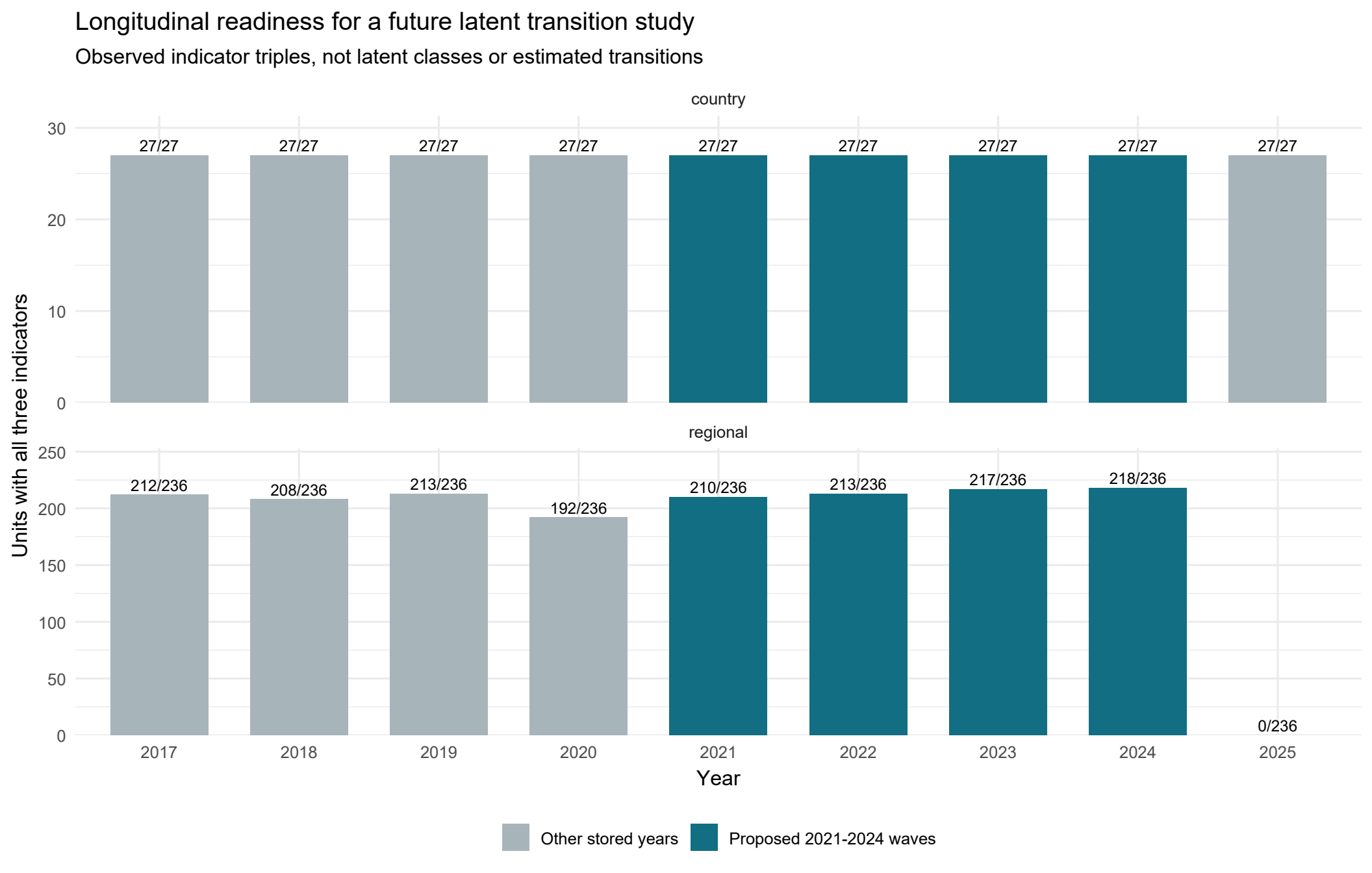

Spatial autocorrelation under alternative coverage choices
asinh(rate), k=4 symmetric union, 9,999 permutations; samples and W differ
EU_all_ages
EU_age_plus_PT
EU_mixed_age
EU_COMPLET
Europe_plus_PT
Europe_all_ages
Europe_without_UK
EU_REACH
n=26; p=0.1938
n=27; p=0.2152
n=26; p=0.2560
n=20; p=0.2620
n=35; p=0.8064
n=34; p=0.7096
n=33; p=0.6788
n=20; p=0.5630
-0.10
-0.05
0.00
0.05
0.10
Moran's I

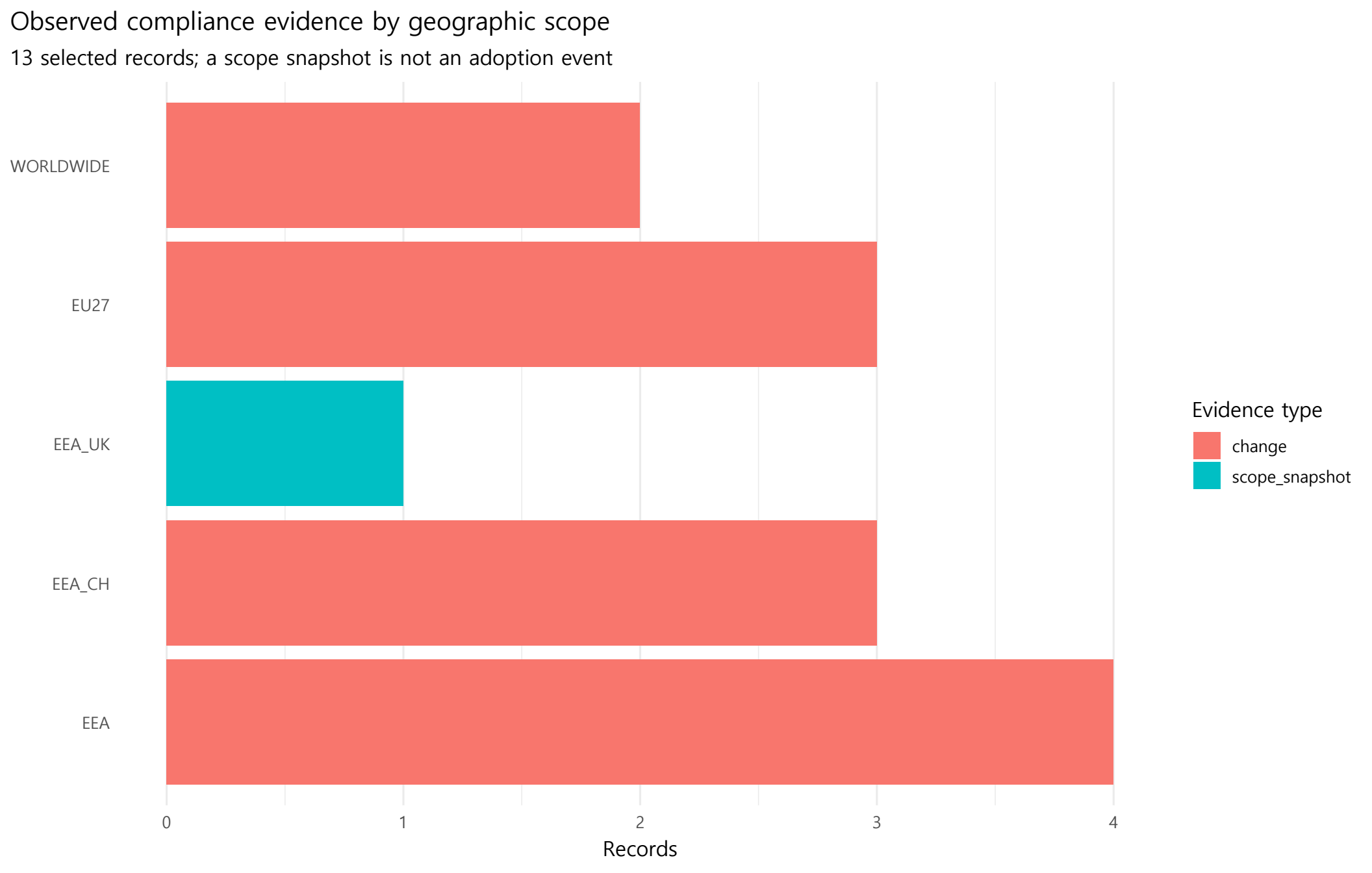
Portugal restored using revised national statistics

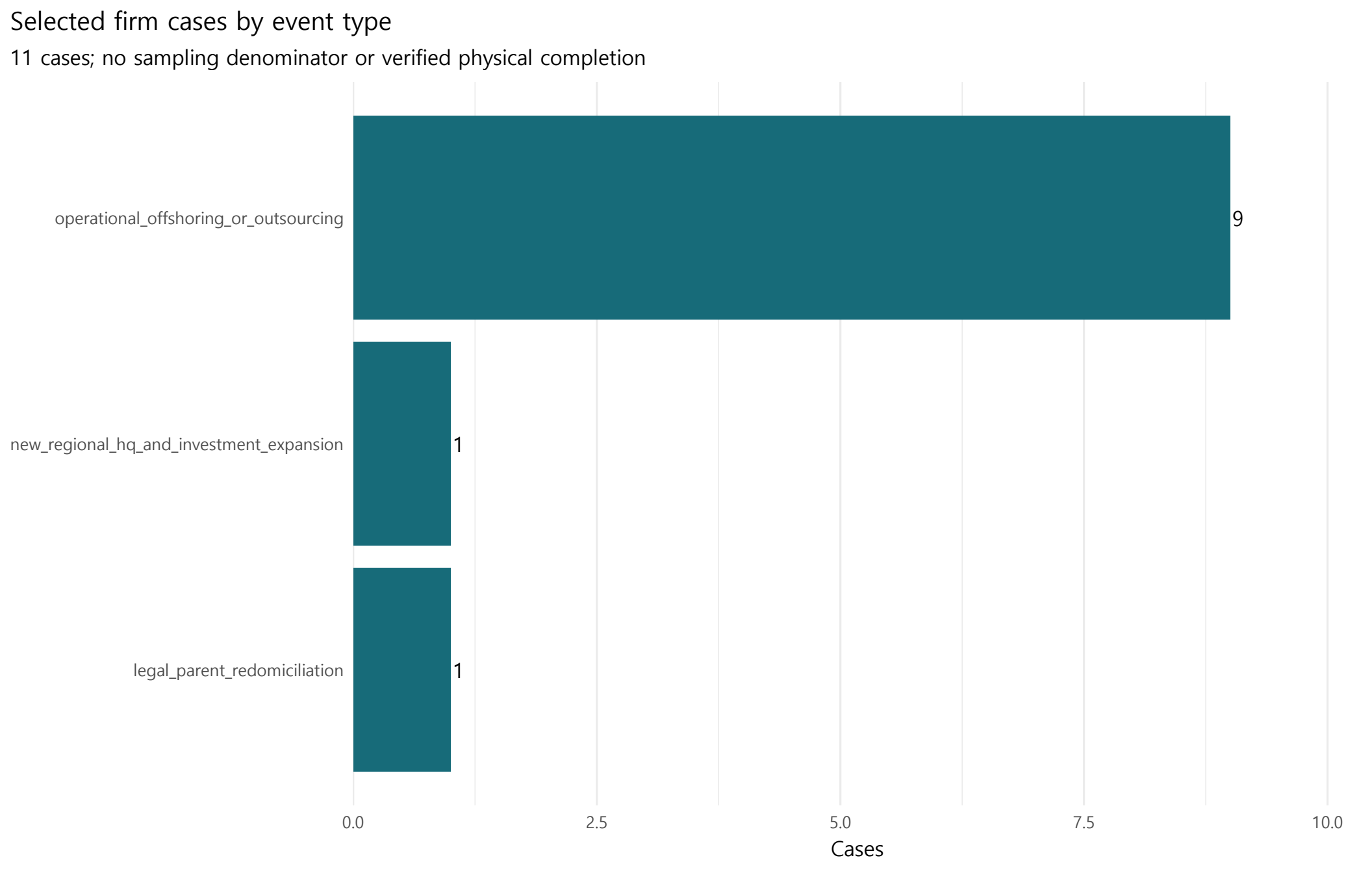
Red outline: Portugal INE numerator and matching revised population series

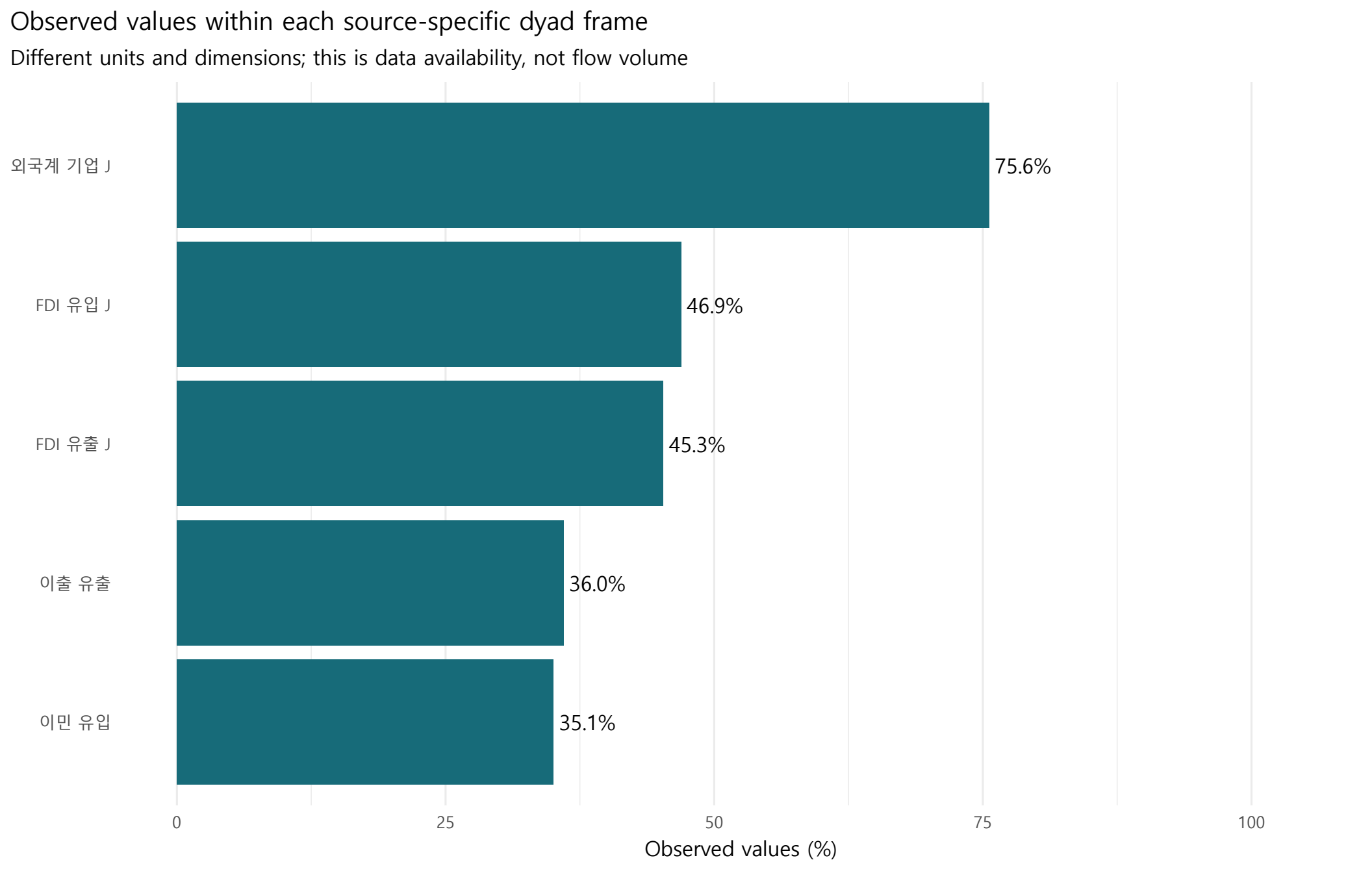


Sensitivity only: COMPLET / REACH / INE age definitions differ. UK appears only in the all-age panel.


Longitudinal readiness for a future latent transition study
Observed indicator triples, not latent classes or estimated transitions
country
regional
Units with all three indicators
Year
27/27
212/236
208/236
213/236
192/236
210/236
213/236
217/236
218/236
0/236
2017
2018
2019
2020
2021
2022
2023
2024
2025
Other stored years
Proposed 2021-2024 waves


Observed compliance evidence by geographic scope
13 selected records; a scope snapshot is not an adoption event
WORLDWIDE
EU27
EEA_UK
EEA_CH
EEA
0
1
2
3
4
Records
Evidence type
change
scope_snapshot


Selected firm cases by event type
11 cases; no sampling denominator or verified physical completion
operational_offshoring_or_outsourcing
9
new_regional_hq_and_investment_expansion
1
legal_parent_redomiciliation
1
0.0
2.5
5.0
7.5
10.0
Cases


Observed values within each source-specific dyad frame
Different units and dimensions; this is data availability, not flow volume
외국계 기업 J
FDI 유입 J
FDI 유출 J
이출 유출
이민 유입
75.6%
46.9%
45.3%
36.0%
35.1%
0
25
50
75
100
Observed values (%)